

Introduction to Single Board Computer (SBC)

What is SBC?

- A Single Board Computer (SBC) is a complete computer built on a single circuit board, with all the necessary components like:
 - Processor (CPU)
 - Memory (RAM)
 - Storage (e.g., SD card or flash)
 - Input/Output ports (USB, HDMI, GPIO, etc.)
 - Networking capabilities (Ethernet, Wi-Fi, Bluetooth, etc.)

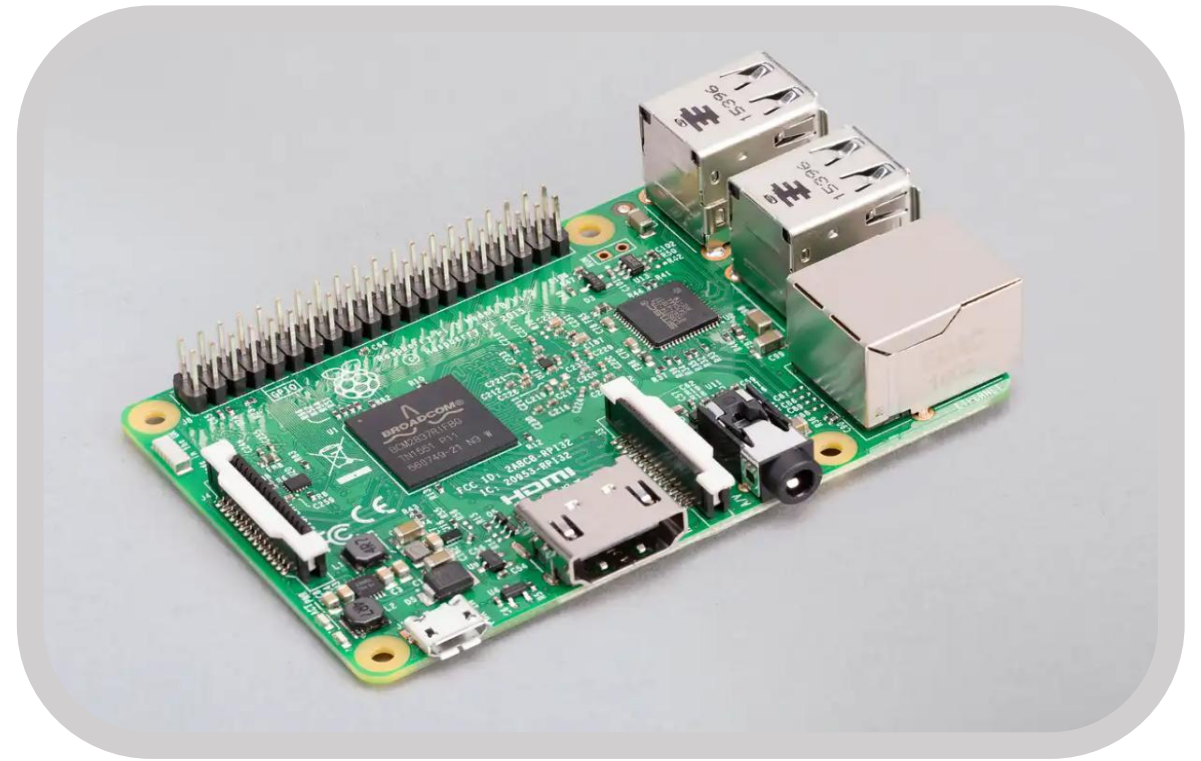
Examples of SBC

- Raspberry Pi
- Beagle Bone Black
- NVIDIA Jetson Nano
- Panda Board
- Friendly ARM
- Phytex Rugged Board
- Many more...

Introduction of Raspberry Pi 3

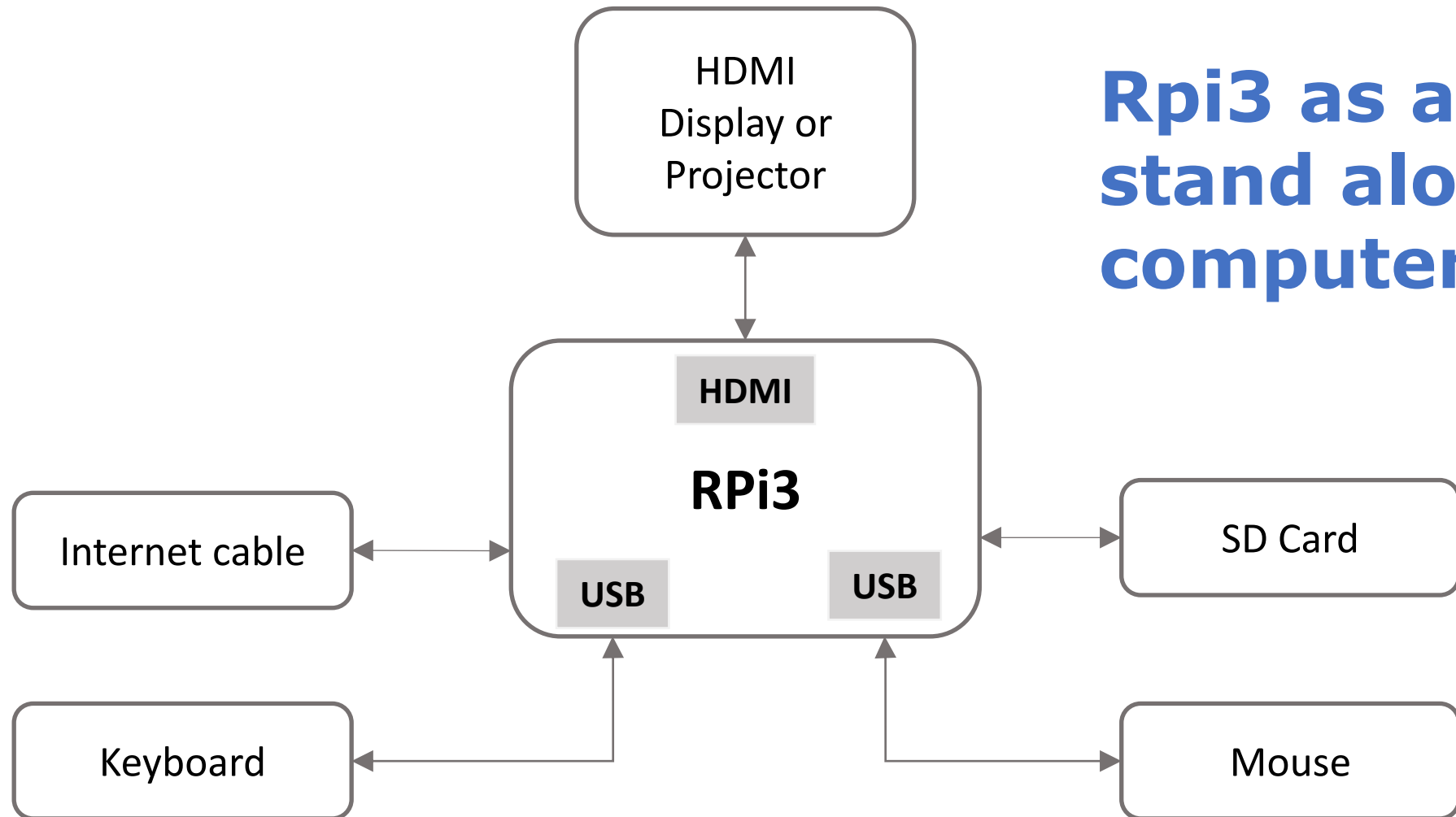
Technical specifications:

- Raspberry Pi 3 Model B is the third-generation Raspberry Pi.
- SBC: RPi3 Model-B
- SoC: BCM2837
- Processor: 1.2GHz Quad-core **ARM Cortex-A53**, 64-bit
- 1GB RAM



- Wireless LAN and Bluetooth Low Energy (BLE) on board
- Ethernet
- 40-pin extended GPIO
- 4 USB ports
- HDMI
- camera port
- display port
- Micro SD port

RPi Connections



**Rpi3 as a
stand alone
computer**

UART Login (CLI login)



SSH Network Login (CLI login)

